

Md Tanzeel Adam Khan

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M.Tech (AI) student at IIT Patna with research in LLM reasoning and search algorithms. 7+ months building production AI systems at Mercedes-Benz R&D India (SSV team). Skilled in multi-agent architectures, model fine-tuning (LoRA/PEFT), and retrieval-augmented generation.

Publication

Beyond Linear Reasoning: A* Search vs Chain-of-Thought for Multi-Strategy LLM Inference

- *Preprint* | First Author | Proposes a budgeted A*-style search framework over LLM reasoning traces with formal admissibility proofs, task-specific heuristics, and weighted goal-node voting. Evaluated on 3 benchmarks (StrategyQA, LogicQA, HotpotQA) across 2 models (Qwen-0.5B, LLaMA-3.1-8B) with 1,500+ paired instances. A* achieves **30.0% vs 8.4% (CoT)** on multi-hop QA; oracle analysis reveals **78.4%** complementarity upper-bound.

Research Experience

Indian Institute of Technology Patna

Jul 2024 – Jun 2026

M.Tech Thesis Research | LLM Reasoning & Search | Advisor: Prof. Asif Ekbal

Patna, India

- Designed a heuristic-guided tree-search algorithm over LLM reasoning traces. Formalized cost functions (path length + inverse confidence), proved admissibility for depth-plus-coverage heuristics, and ran full paired evaluations with 95% Wilson confidence intervals across 3 benchmarks and 1,500+ evaluation instances.
- Trained LoRA adapters (rank-16, PEFT) on Qwen-0.5B and LLaMA-3.1-8B to distill successful A* search traces into linear CoT rationales. Built end-to-end pipeline: rationale rewriting, quality filtering, and supervised fine-tuning with HuggingFace Trainer. Achieved **33% rescue rate** on CoT failure cases with **50% preservation** of existing correct answers on LLaMA-3.1-8B.
- Designed 4-bucket evaluation framework (both-correct / CoT-only / A*-only / both-wrong), multi-run majority-vote stability analysis (3 runs × 93 questions), and instance-level complementarity decomposition. Conducted ablation studies across learning rates, epoch schedules, and data mixing ratios.

Professional Experience

Mercedes-Benz Research & Development India (MBRDI)

Nov 2025 – Present

AI Engineer Intern | AMG System Software Validation Team

Bengaluru, India

- Designed and built an AI-powered root-cause analysis platform (FAST) using a LangGraph ReAct agent with 14 domain-specific tools, performing full-depth BFS over automotive signal dependency graphs (94,523 nodes). Achieved 100% upstream resolution accuracy (10/10 benchmark signals across 3 ECU modules). Integrated Gemini 2.5 Pro, GPT-5, and Claude backends with hybrid RAG (dense + BM25 + Reciprocal Rank Fusion).
- Deployed a production RAG system with hybrid BM25 + ANN retrieval, cross-encoder reranking (ms-marco-MiniLM), and LangGraph state-machine orchestration for the validation team. Achieved **77% answer rate** on a 100-query evaluation benchmark, 100% out-of-domain guardrail compliance, and **5.5× latency reduction** (22s → 4s) through semantic caching, connection pooling, and async parallelism.
- Architected a dual-agent system (A2A) for automated ASPICE SWE.6 compliance assessment across 60+ evaluation dimensions. Implemented evidence-grounded evaluation requiring source citation at every step, hierarchical scoring aggregation, and multimodal document ingestion supporting 7 formats.

Moshi Moshi

Jun 2025 – Aug 2025

AI Engineer Intern

Bengaluru, India

- Built GenAI tools for automated website content generation (35% workflow efficiency gain) and LLM-driven personalized client outreach with automated email extraction (25% turnaround improvement).

Technical Skills

ML & Training: PyTorch, LoRA/QLoRA, PEFT, HuggingFace Transformers/Trainer, Supervised Fine-Tuning, Ablation Studies

LLM Systems: LangGraph, LangChain, RAG, Multi-Agent Systems (A2A/ReAct), Tool-Use Agents, Prompt Engineering

Models: LLaMA-3.1-8B, Qwen-0.5B/2.5, Claude, Gemini 2.5 Pro, GPT-5, Ollama, vLLM

Retrieval & Search: A* Search, BM25, ANN, Cross-Encoders, Reciprocal Rank Fusion, ChromaDB, LanceDB, Sentence-Transformers

Engineering: Python, SQL, NumPy, Pandas, SciPy, NetworkX, scikit-learn, FastAPI, Streamlit, Docker, Git, asyncio

Research Methods: Experimental Design, Paired Evaluation, Confidence Intervals, Heuristic Design, Admissibility Proofs

Education

Indian Institute of Technology Patna

2024 – 2026

Master of Technology in Artificial Intelligence (CGPA: 8.5/10)

Patna, India

Haldia Institute of Technology

2020 – 2024

Bachelor of Technology in CSE – AI & ML (CGPA: 8.9/10)

Haldia, India